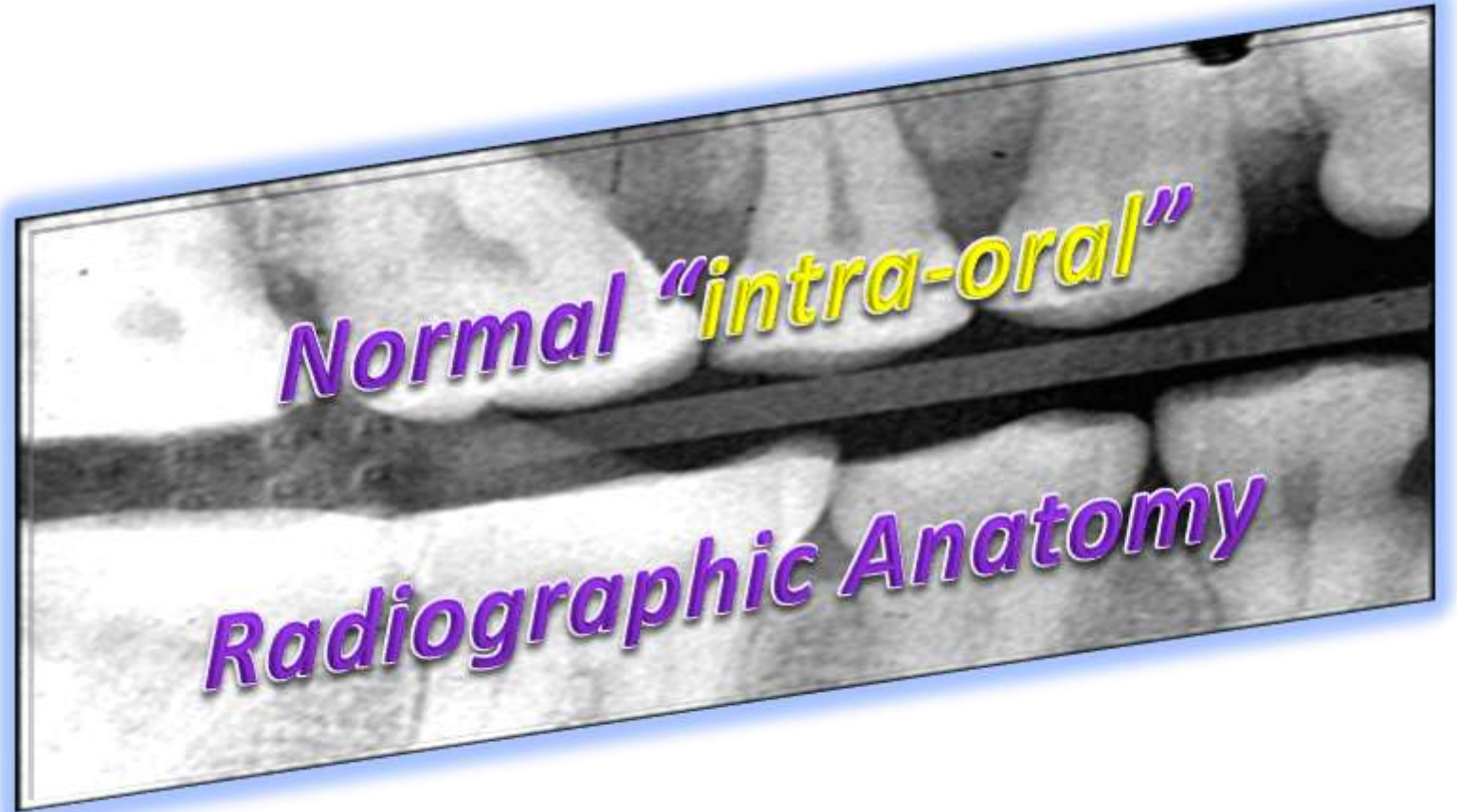




جامعة الكفيل
University of Alkafeel





- * **Radiolucent:** Radiolucent refers to that portion of a processed radiograph that is **dark or black**. A structure that appears radiolucent on a radiograph lacks density and permits the passage of the X-ray beam with little or no resistance.





- **Radiopaque:** Radiopaque refers to that portion of a processed radiograph that appears **light or white**. Radiopaque structures are dense and absorb or resist the passage of the X-ray beam.





- In the oral cavity the relative densities of various natural structures, in order of decreasing density, are enamel, dentin, and cementum, bone, soft tissue, fat, and air.
- Metallic objects, such as restorations, are far denser and hence better absorbers, than enamel.





Normal radiographic anatomy

1. Enamel, Dentin, Cementum and bone :

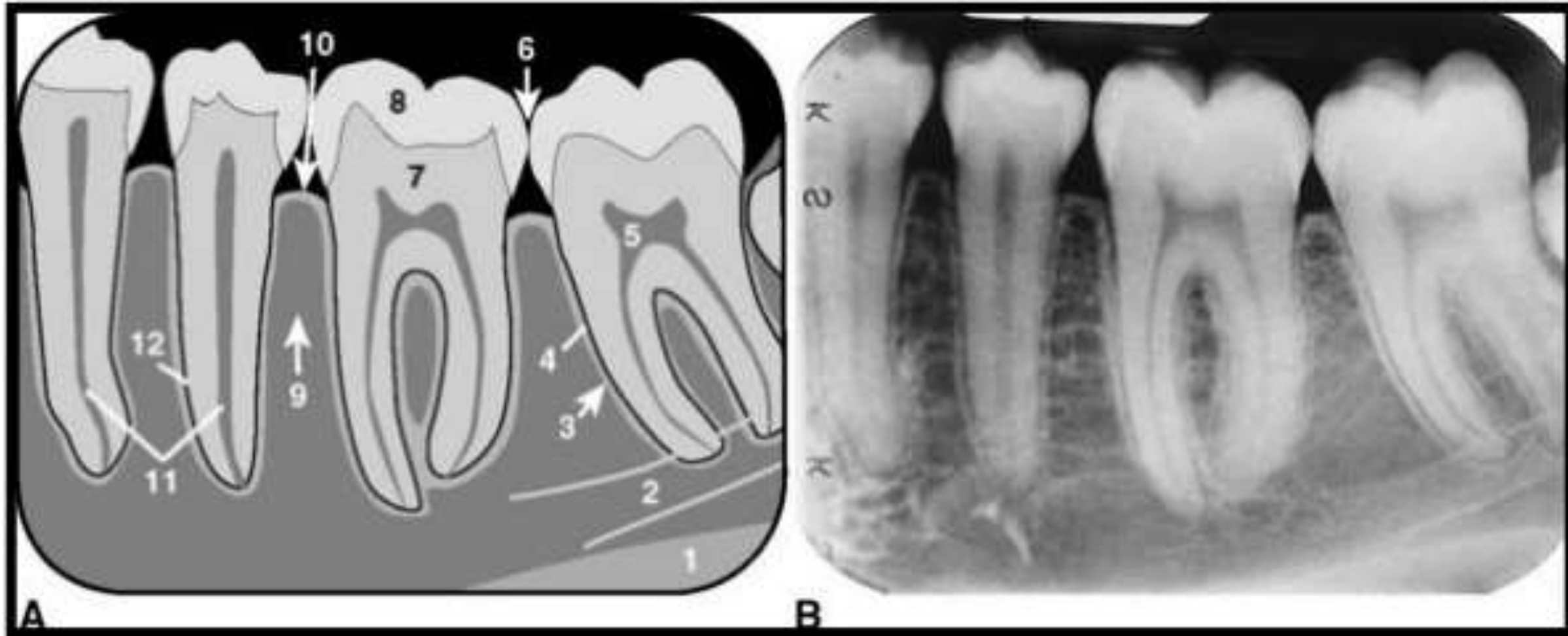
- Enamel (90% mineral content): is the **most radiopaque structure**.
- Dentin (75% mineral content): **less radiopaque than enamel, has the same radiopacity as bone** .
- Cementum (50% mineral content): in most areas of the root surface cannot be seen radiographically. It has the **same density as bone or dentin** .

2. pulp and periodontal ligament : radiolucent

In the healthy periodontium, the PDL space appears as a continuous radiolucent line of uniform thickness.

3. Lamina dura : is the radiopaque border of bone surrounding the periodontal ligament.

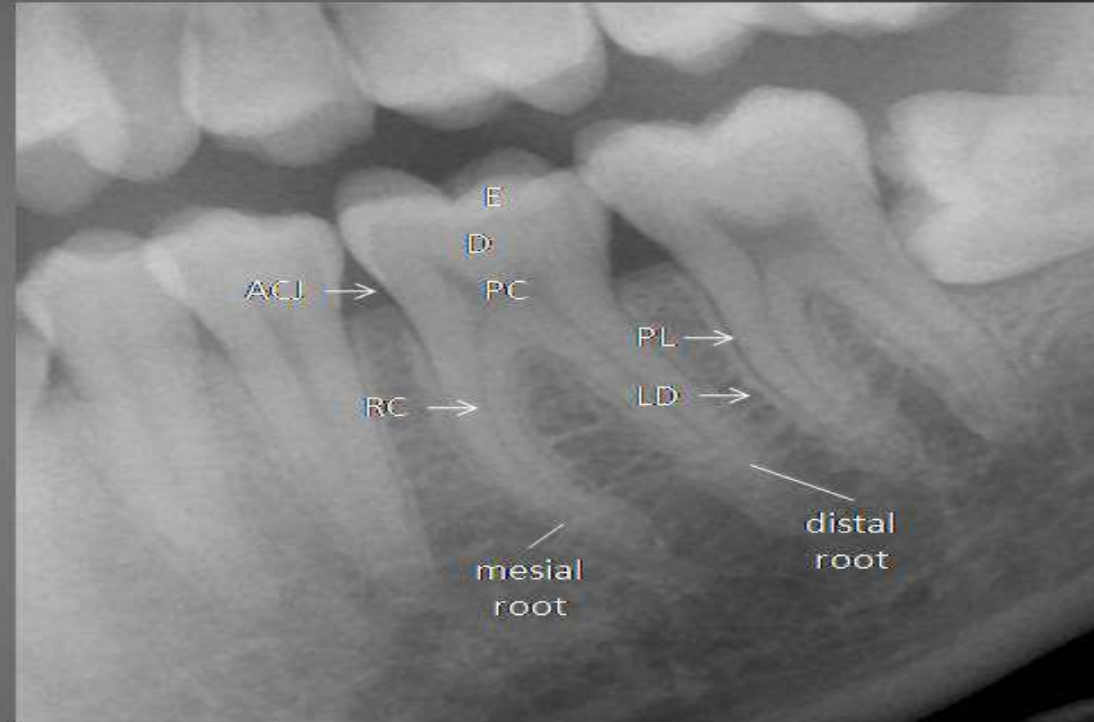






Dental Anatomy

E	enamel
D	dentine
PC	pulp cavity
RC	root canal
ACJ	amelo-cemental junction
PL	periodontal ligament
LD	lamina dura





The composition of bone in the human body can be described

as either cortical or cancellous

Cortical bone, also referred to as compact bone, is the dense outer layer of bone Cancellous bone (also called trabecular bone) is the soft, spongy bone located between two layers of dense cortical bone





trabeculae in the anterior maxilla are typically thin and numerous in the posterior maxilla the trabecular pattern is usually quite similar to that in the anterior maxilla, although the marrow spaces may be slightly larger.





In the anterior mandible the trabeculae are somewhat thicker than in the maxilla, resulting in a coarser pattern, with trabecular plates that are oriented more horizontally

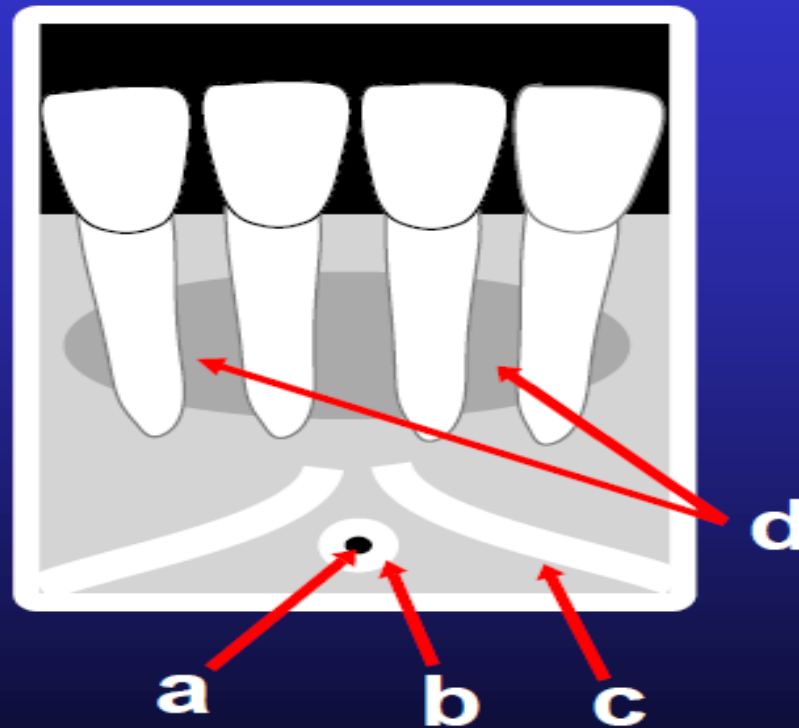
In the posterior mandible the periradicular trabeculae and marrow spaces may **be comparable to those in the anterior mandible** but are usually somewhat larger





Landmarks in radiographs of the mandible

Mandibular Incisor

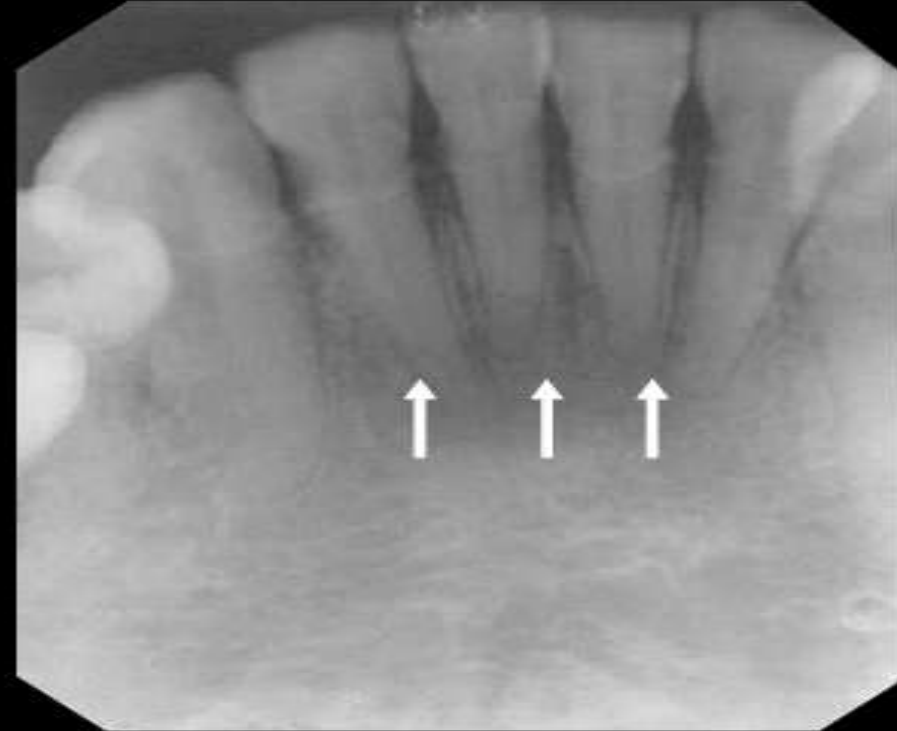
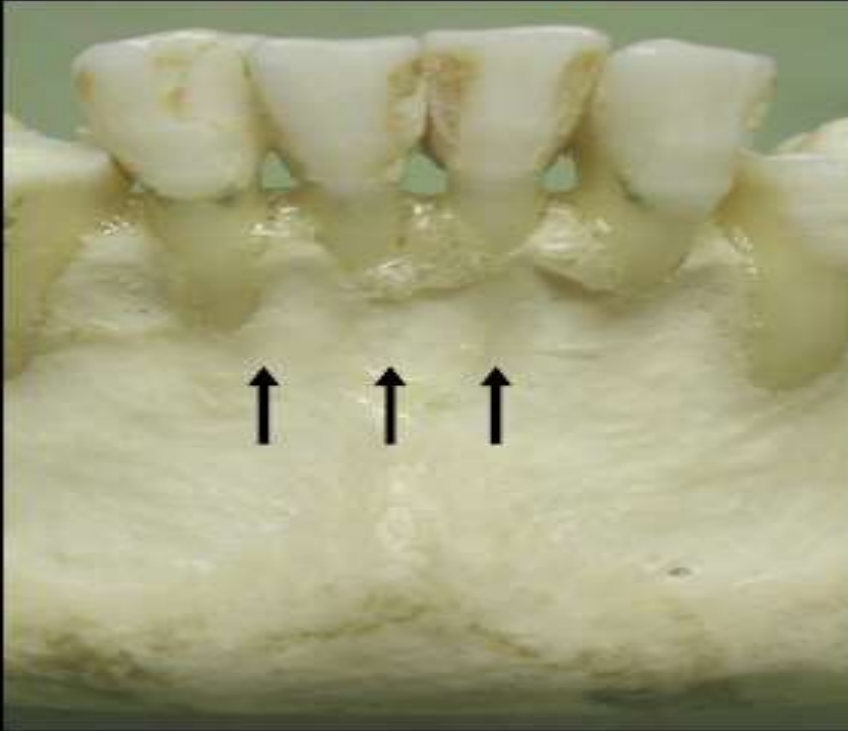


- a. lingual foramen
- b. genial tubercles
- c. mental ridge
- d. mental fossa





facial view

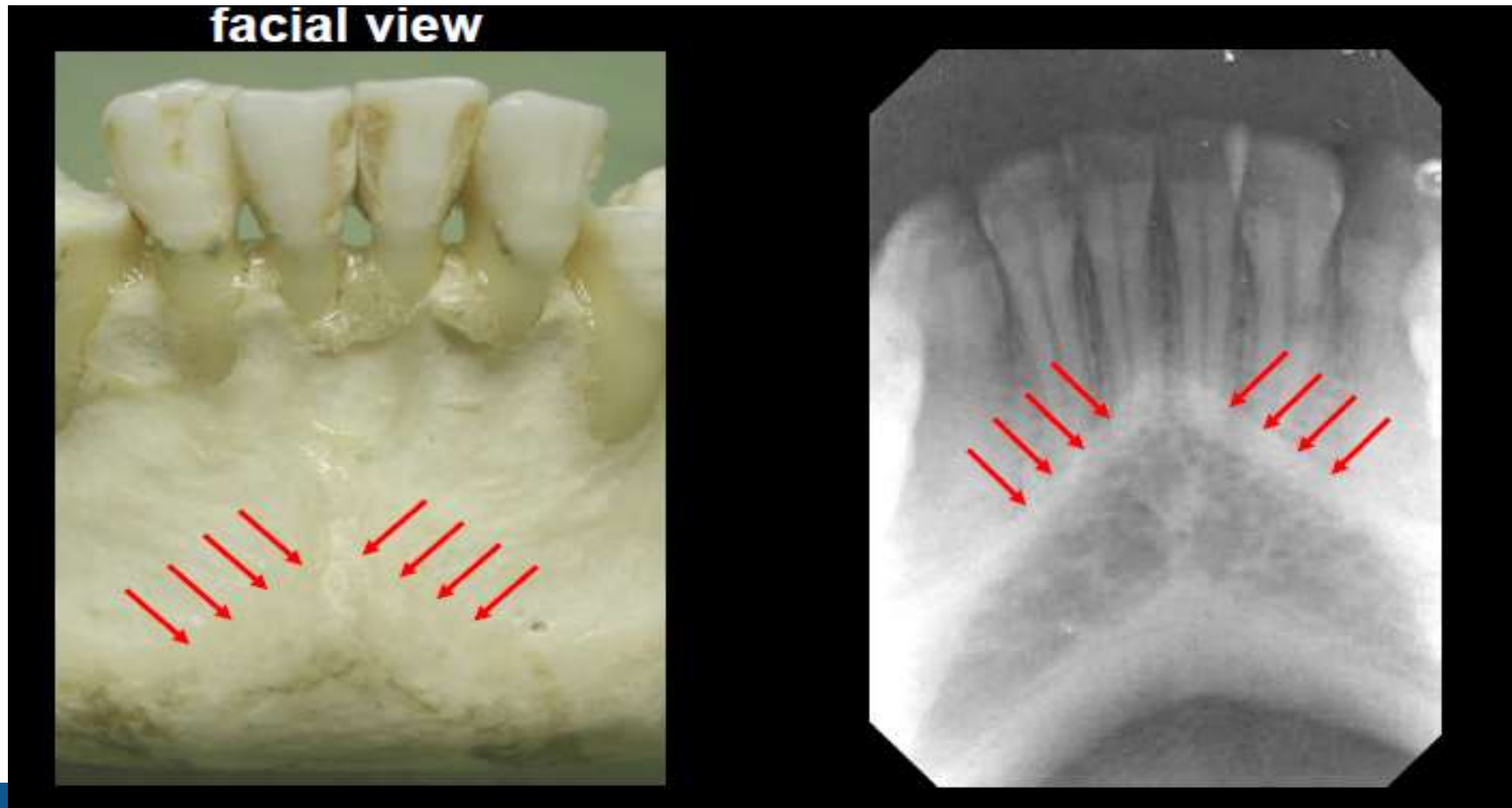


Mental fossa. This represents a depression on the labial aspect of the mandible overlying the roots of the incisors. The resulting radiolucency may be mistaken for pathology.



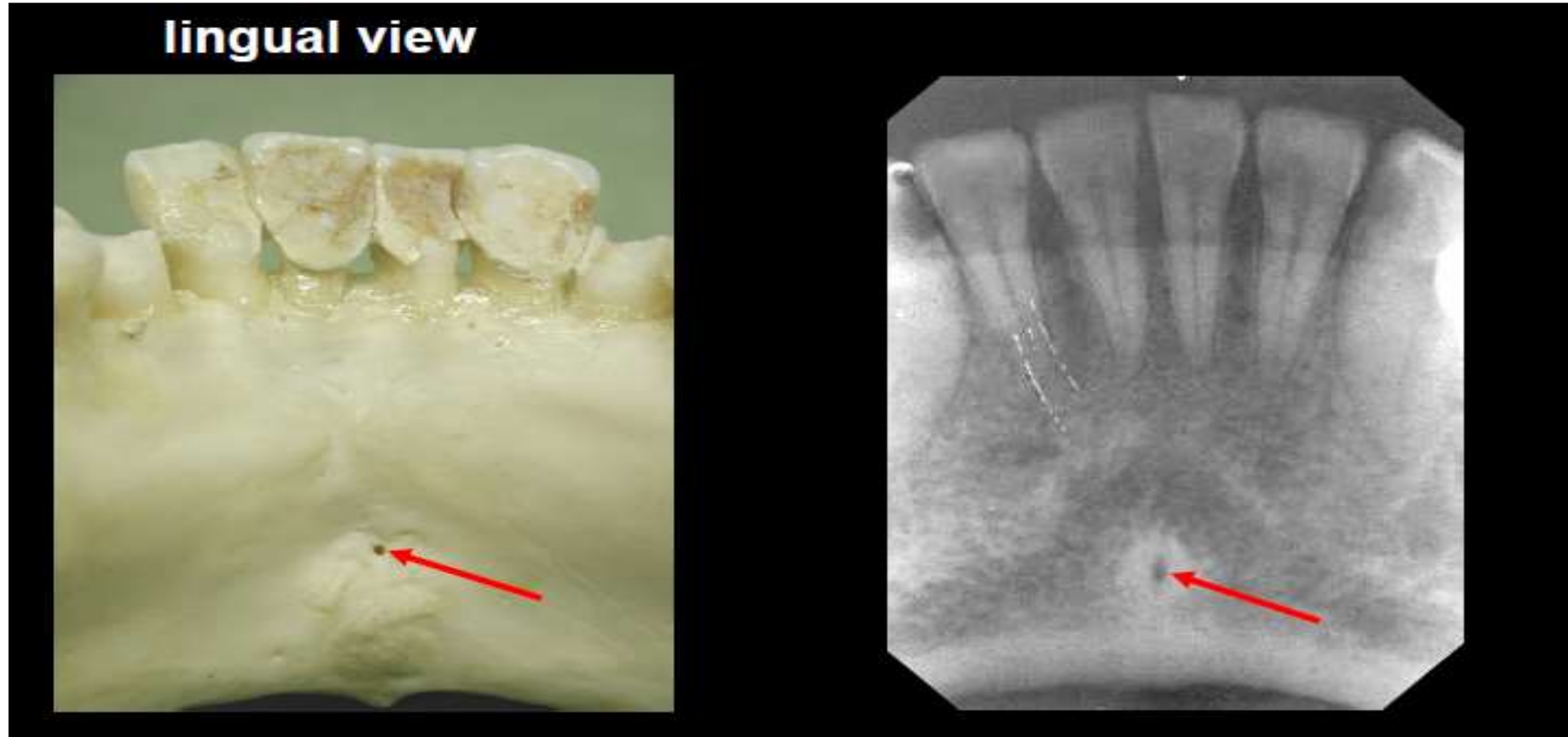


Mental ridge : bony prominence extending from midline of the mandible to the area of premolars (Radiopaque).



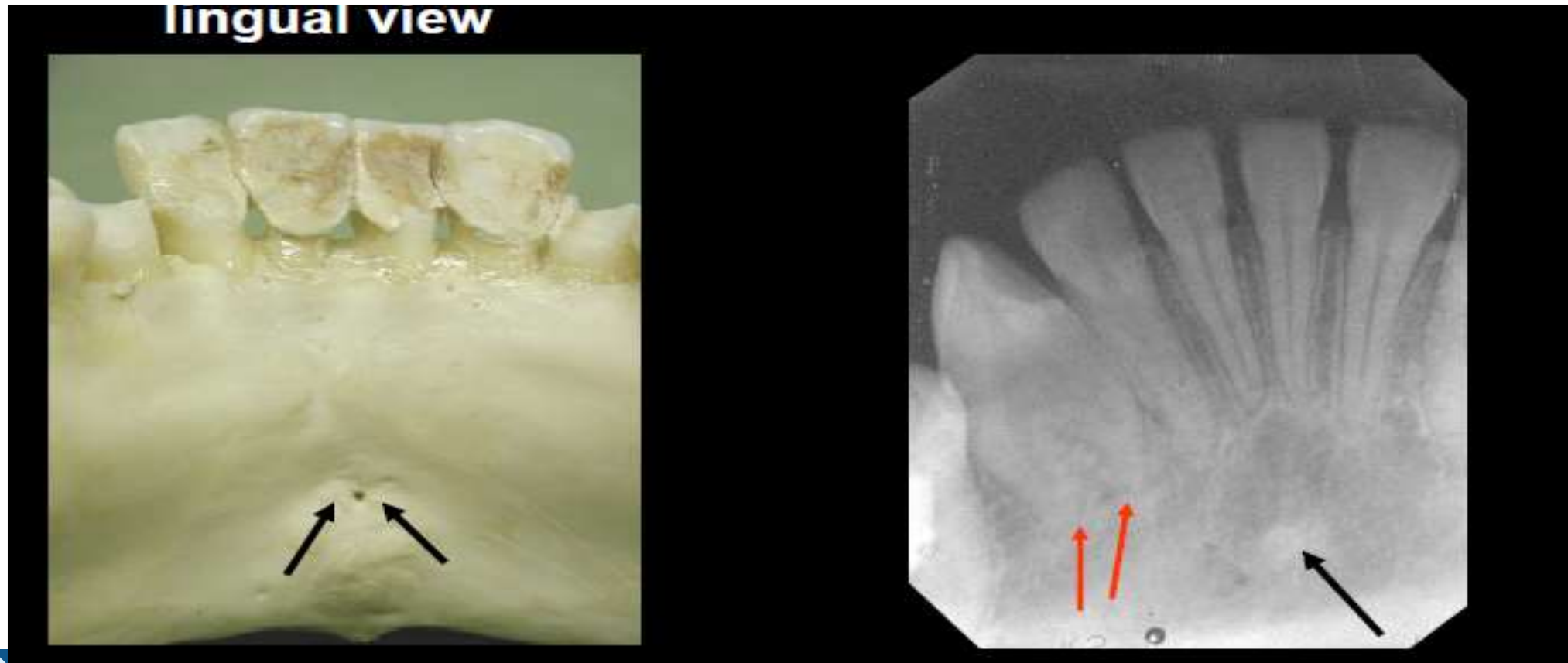


Lingual foramen : single radiolucent area in the midline of the mandible.



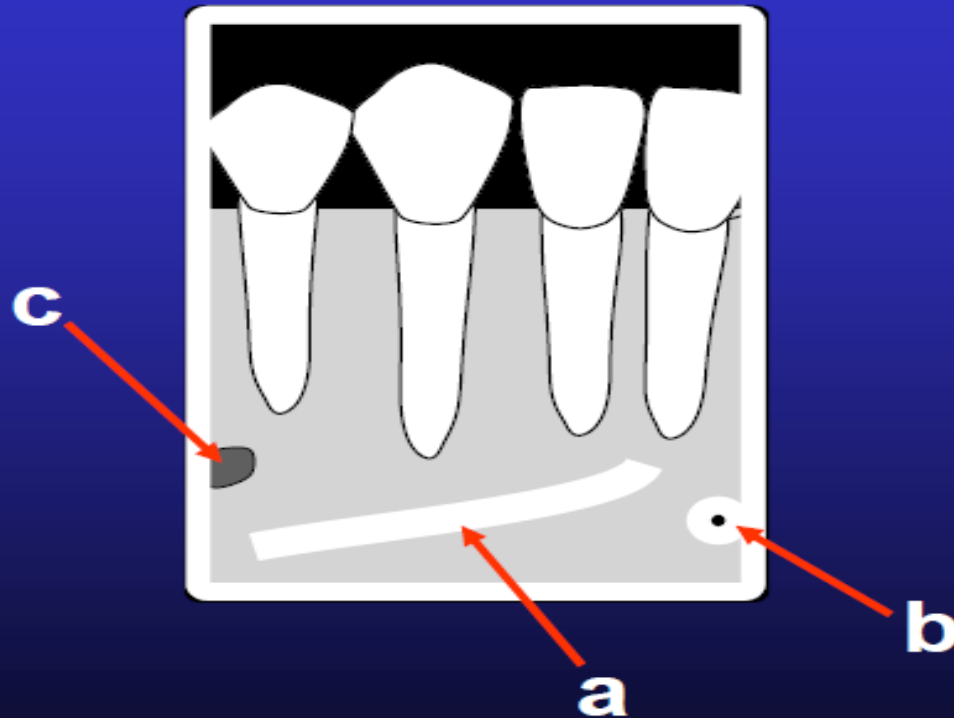


Genial tubercles : radiopaque zones located in the radiographs of lower incisors area





Mandibular Canine

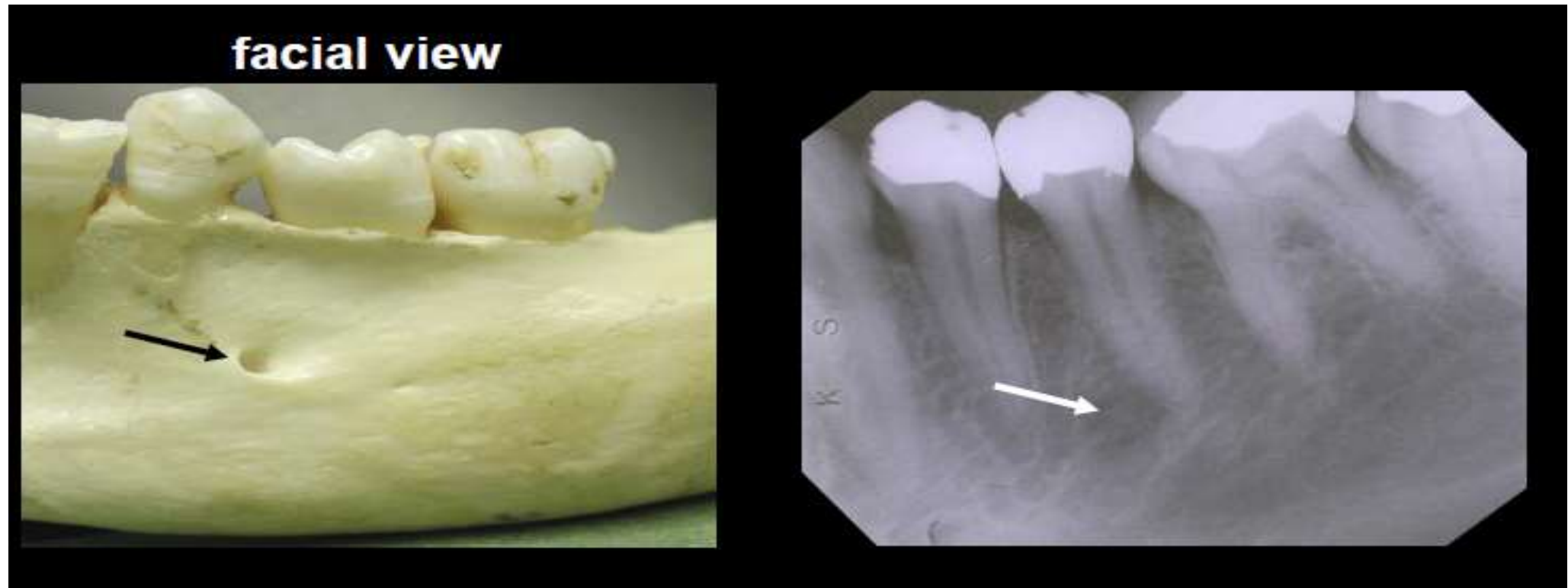


a = mental ridge
**b = genial tubercles/
lingual foramen**
c = mental foramen



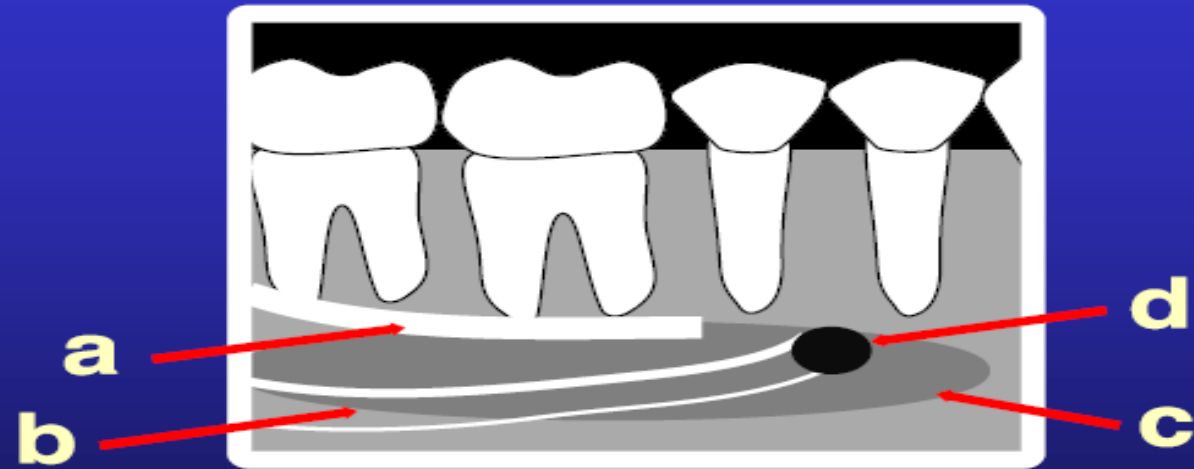


Mental foramen : oval, round, or irregular radiolucency in the radiographs of premolars, when superimposed of the apex of a tooth it may misdiagnosed as periapical pathology.





Mandibular Premolar

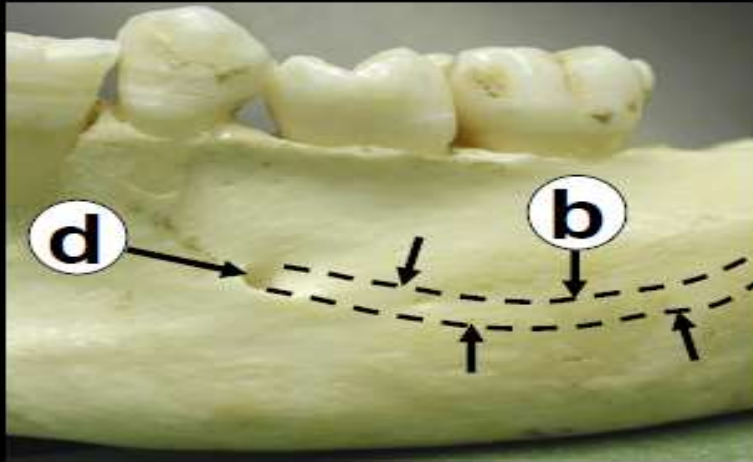


a = mylohyoid ridge
b = mandibular canal
c = submandibular gland fossa
d = mental foramen



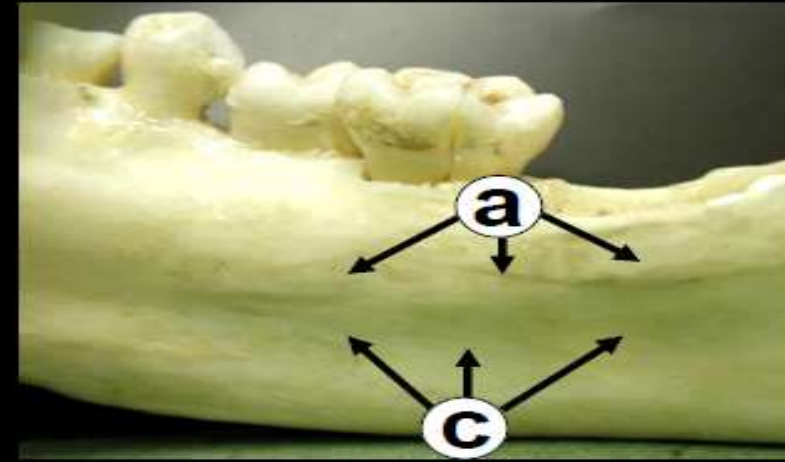


facial view



b = mandibular canal
d = mental foramen

lingual view

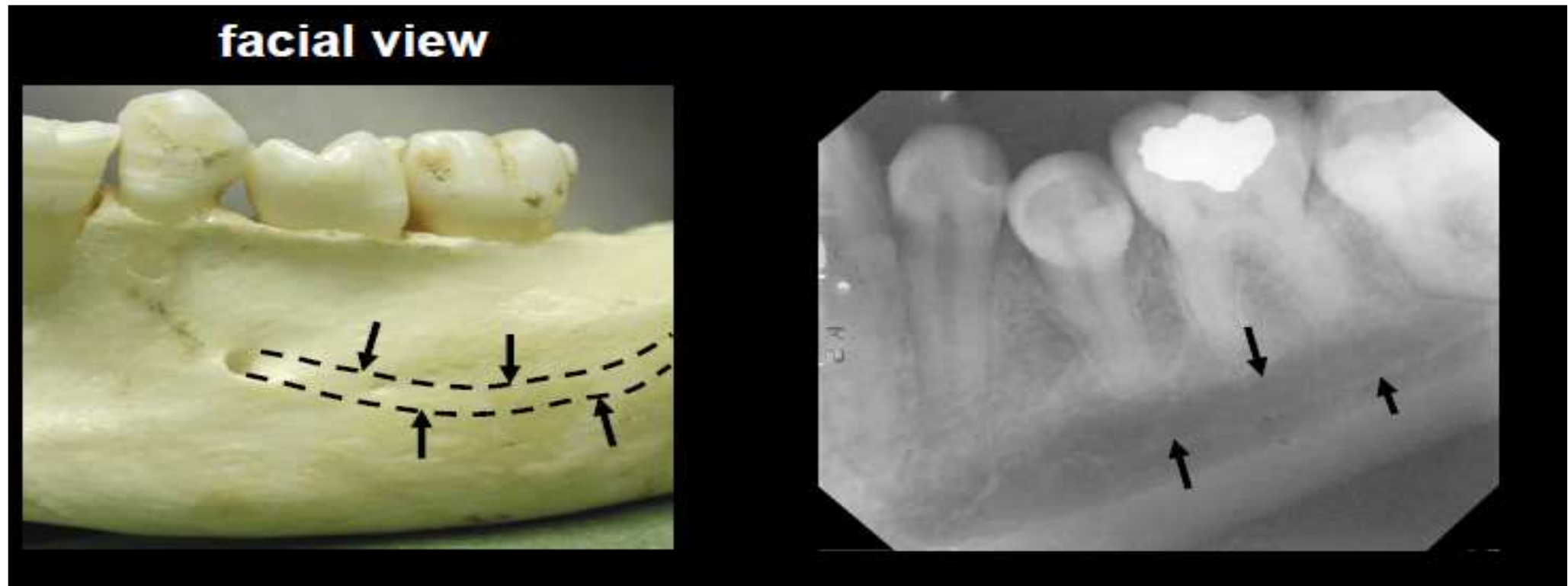


**a = mylohyoid ridge
(internal oblique)**
**c = submandibular gland
fossa**



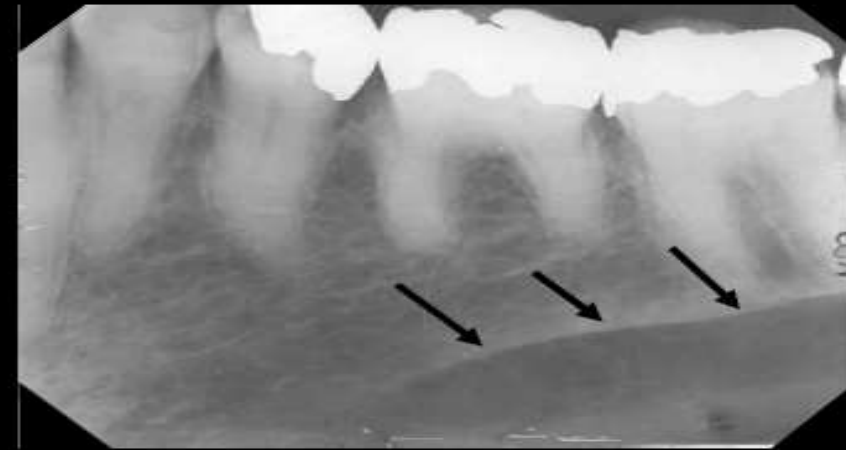
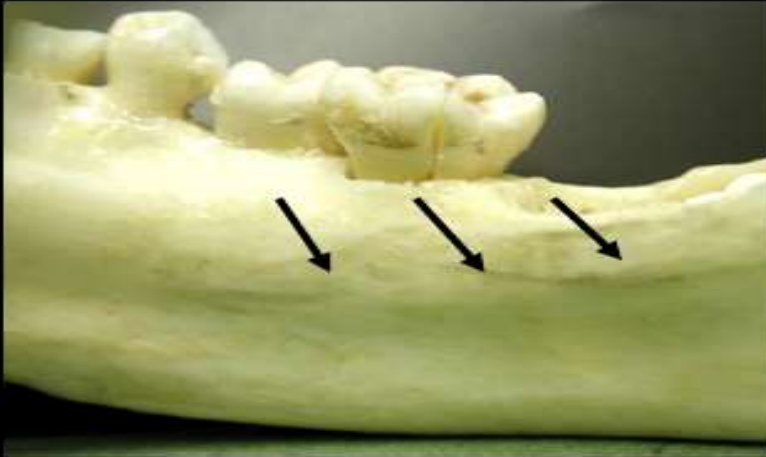


Mandibular canal (inferior dental canal (ID)) radiolucent structure extends from the mental foramen to the mandibular foramen.





lingual view

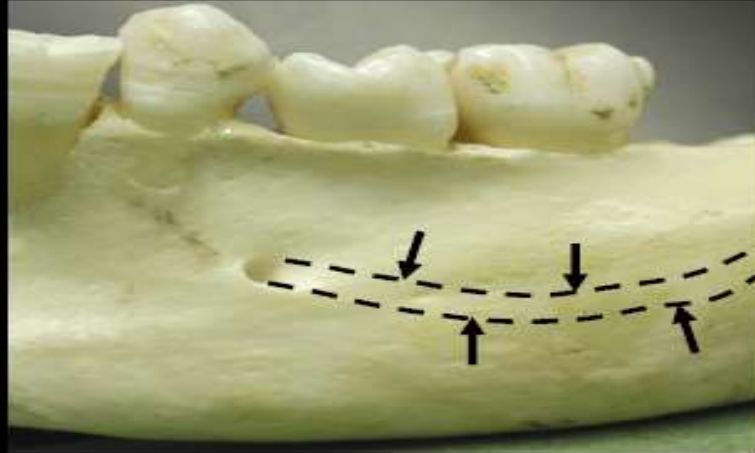


Mylohyoid (internal oblique) ridge. This radiopaque ridge is the attachment for the mylohyoid muscle. The ridge runs downward and forward from the third molar region to the area of the premolars.





facial view

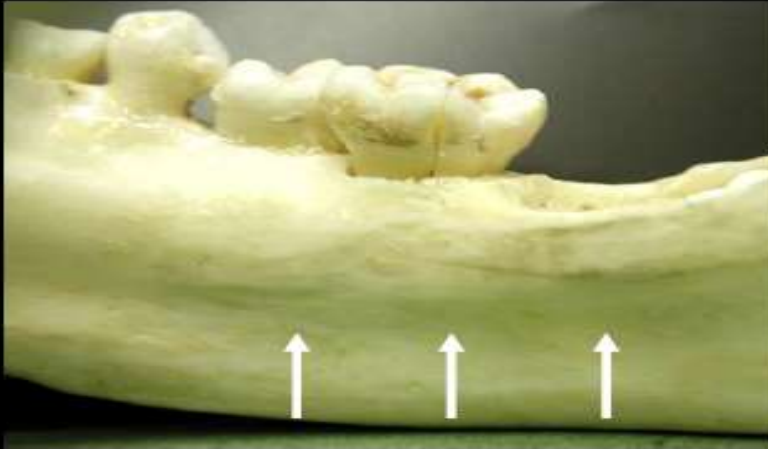


Mandibular canal. (Inferior alveolar canal). Runs downward from the mandibular foramen to the mental foramen, passing close to the roots of the molars. More easily seen in the molar periapical.





lingual view

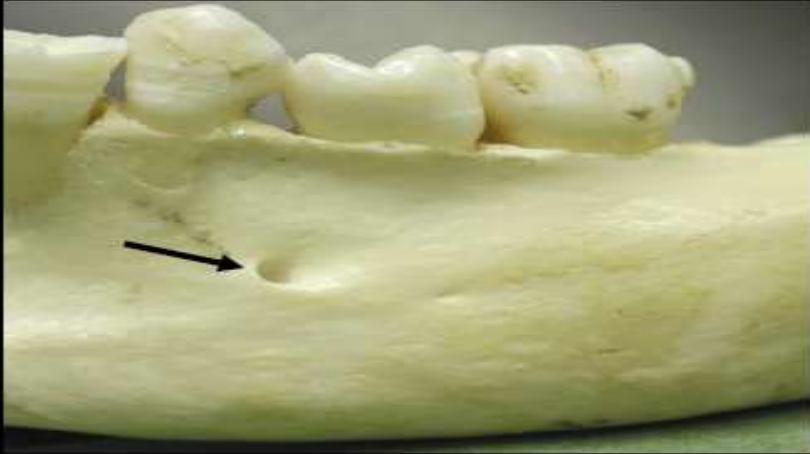


Submandibular gland fossa. The depression below the mylohyoid ridge where the submandibular gland is located. More obvious in the molar periapical film.





facial view

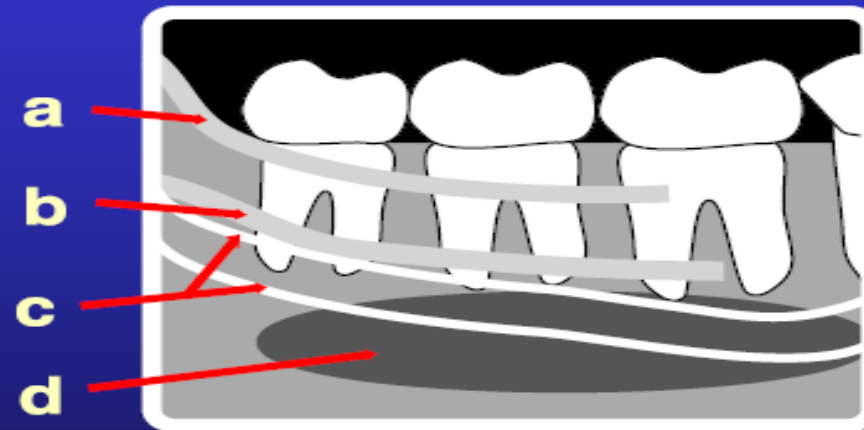


Mental foramen. Usually located midway between the upper and lower borders of the body of the mandible, in the area of the premolars. May mimic pathology if superimposed over the apex of one of the premolars.





Mandibular Molar

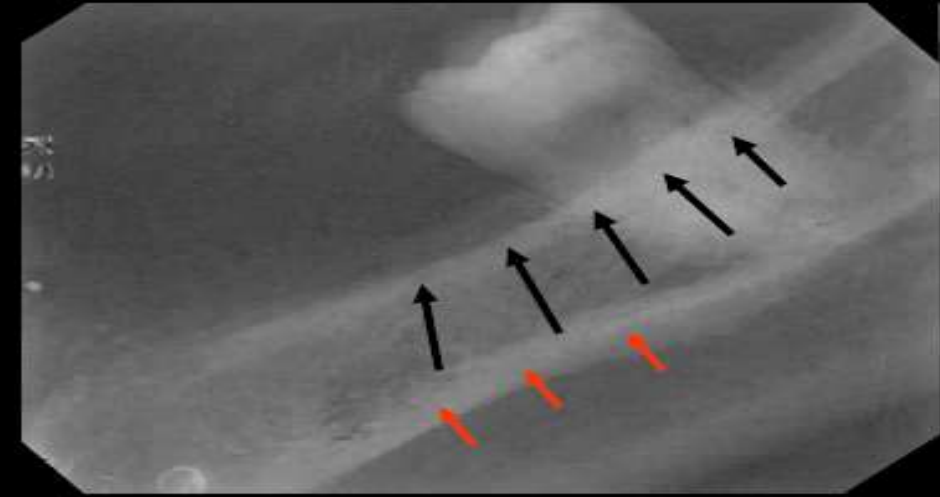
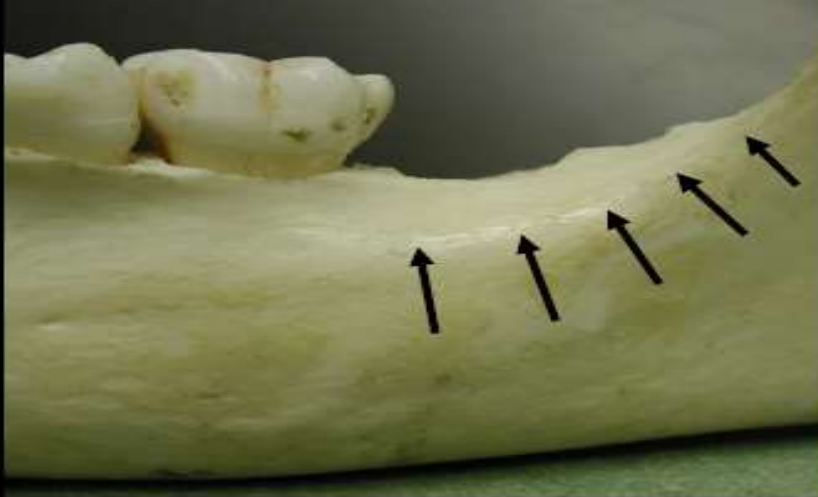


a = external oblique ridge
b = mylohyoid ridge
c = mandibular canal
d = submandibular gland fossa





facial view

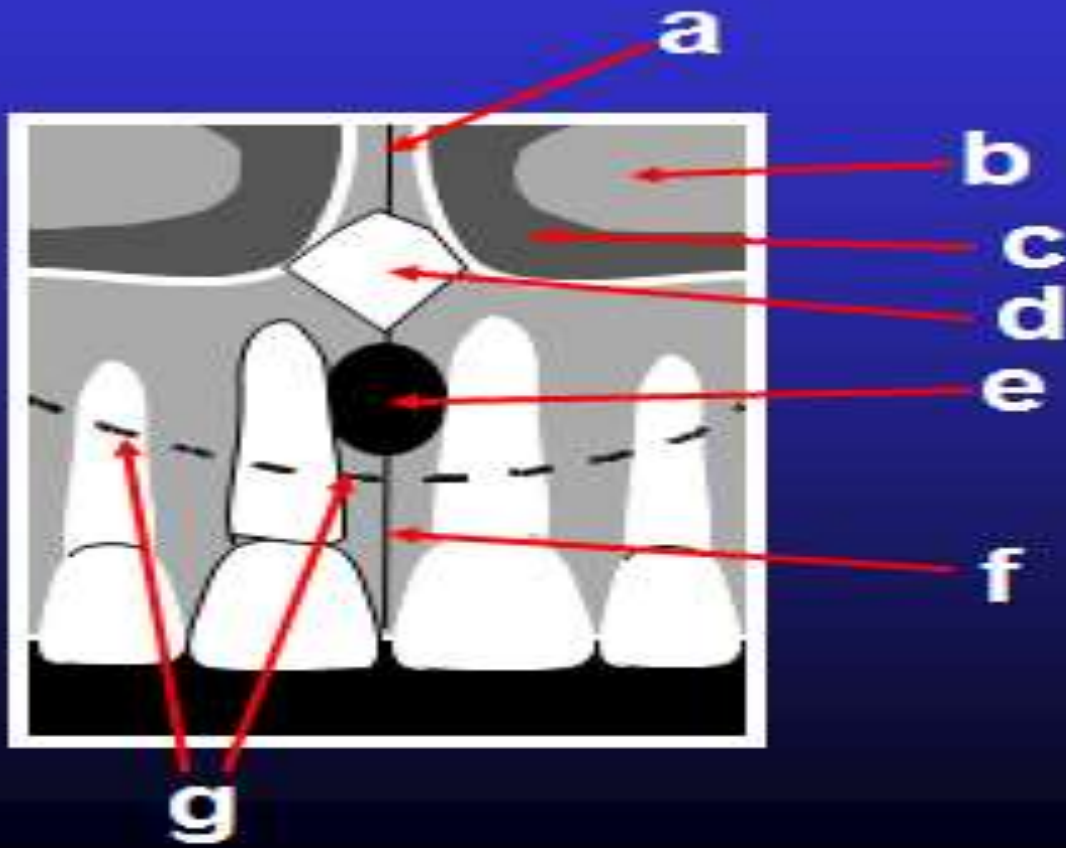


External oblique ridge. A continuation of the anterior border of the ramus, passing downward and forward on the buccal side of the mandible. It appears as a distinct radiopaque line which usually ends anteriorly in the area of the first molar. Serves as an attachment of the buccinator muscle. (The red arrows point to the mylohyoid ridge).





Maxillary Incisor

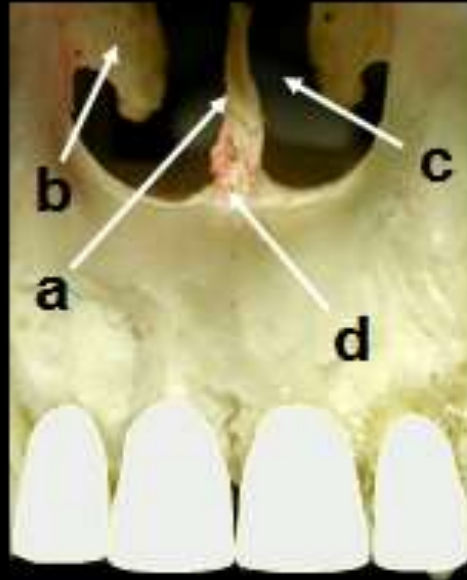


- a = nasal septum
- b = inferior concha
- c = nasal fossa
- d = anterior nasal spine
- e = incisive foramen
- f = median palatal suture
- g = soft tissue of nose



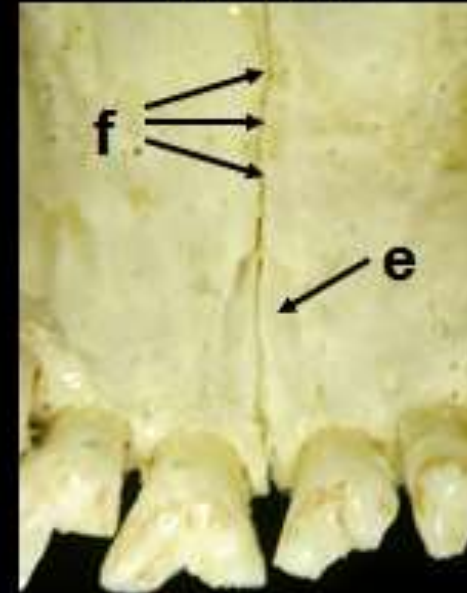


facial view



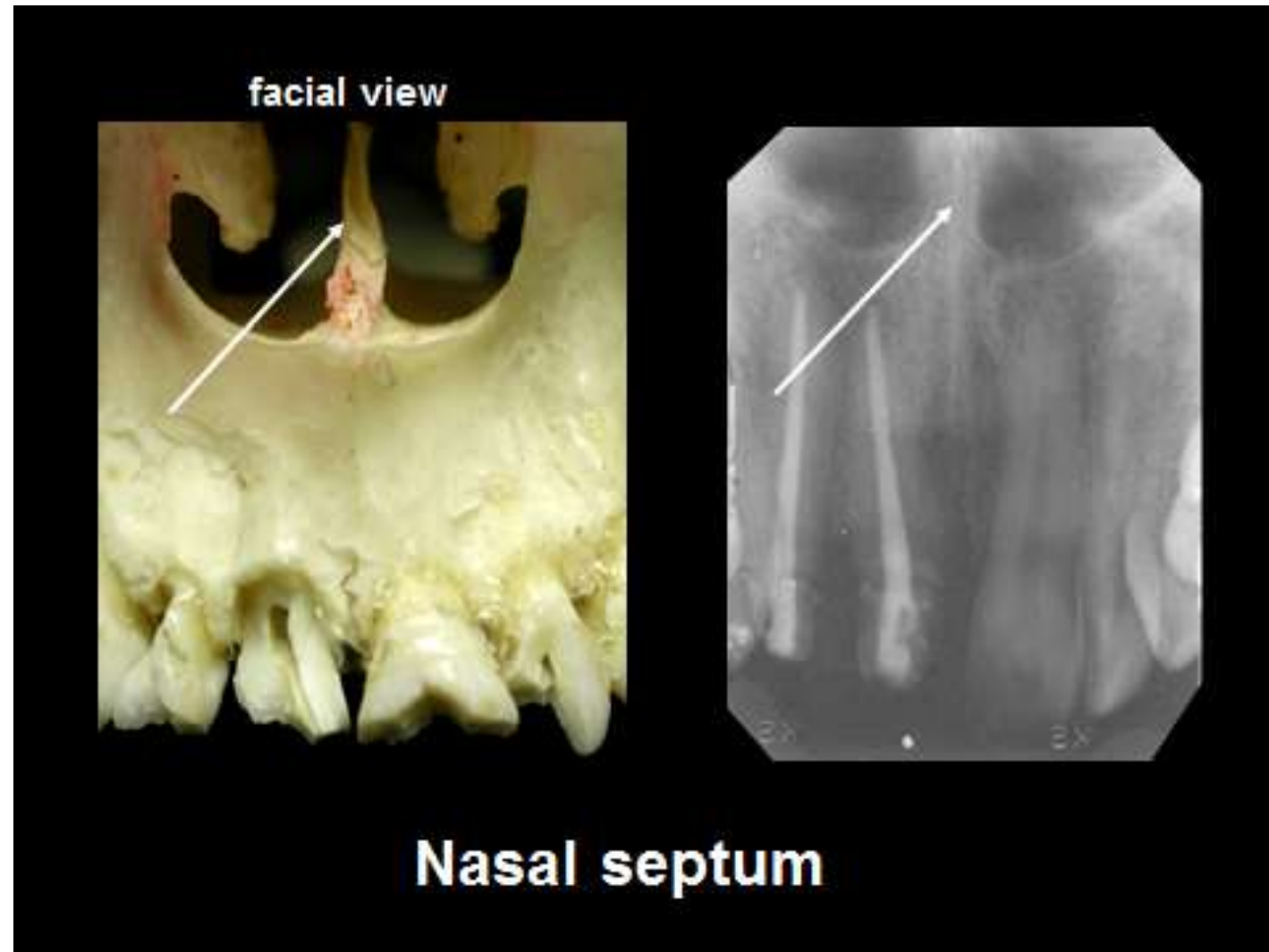
a = nasal septum
b = inferior concha
c = nasal fossa
d = anterior nasal spine

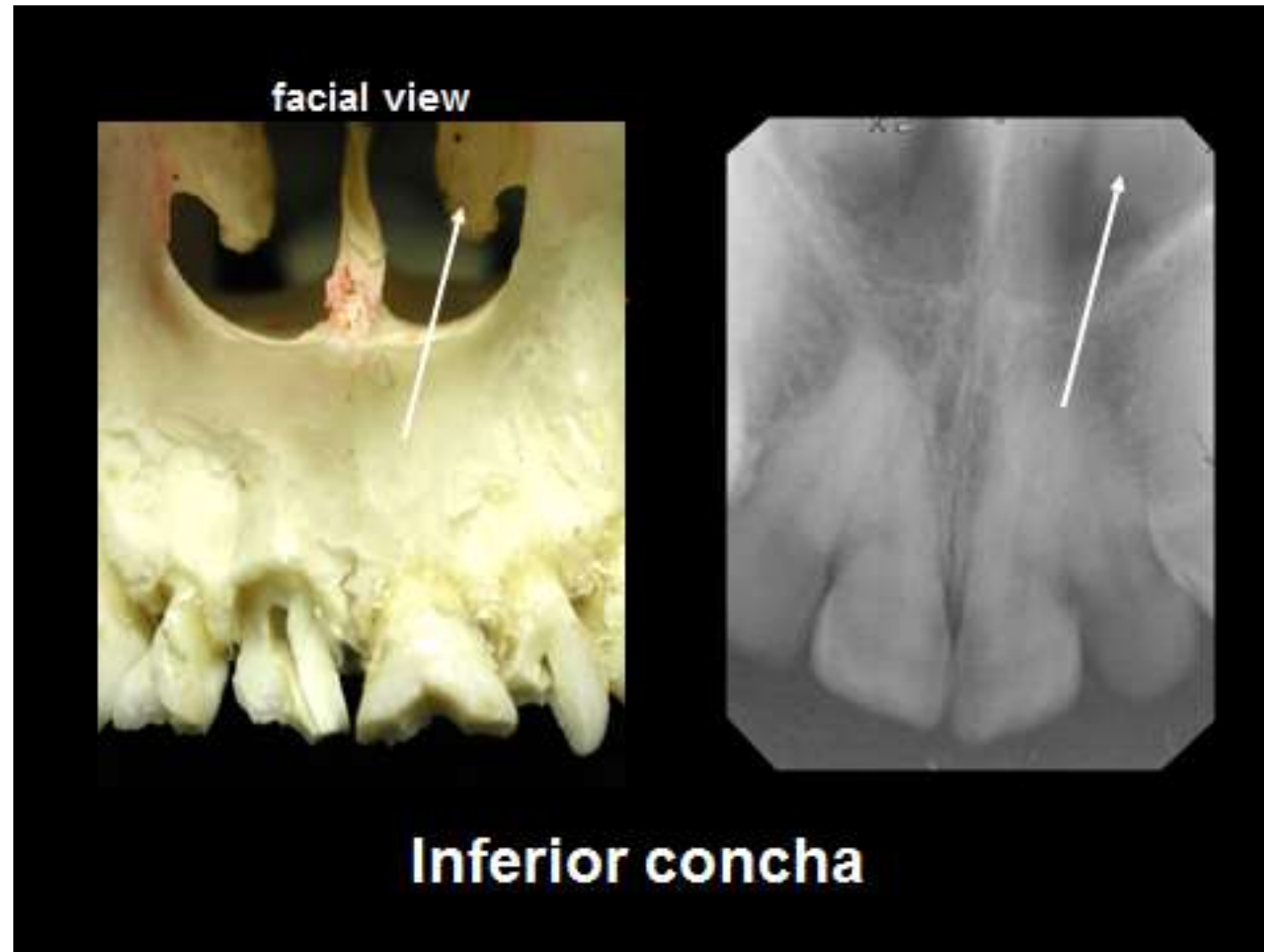
palatal view

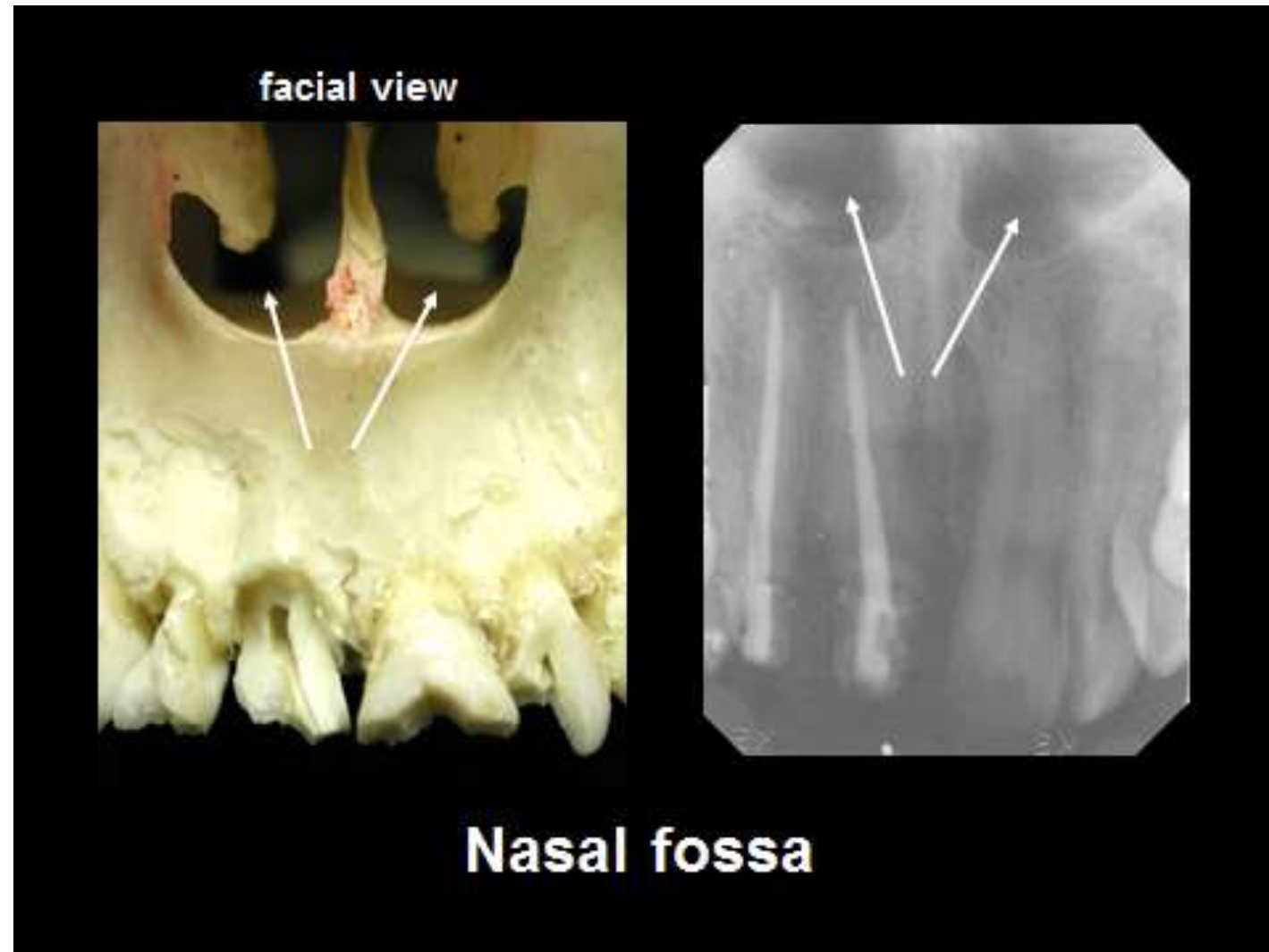


e = incisive foramen
f = median palatal suture



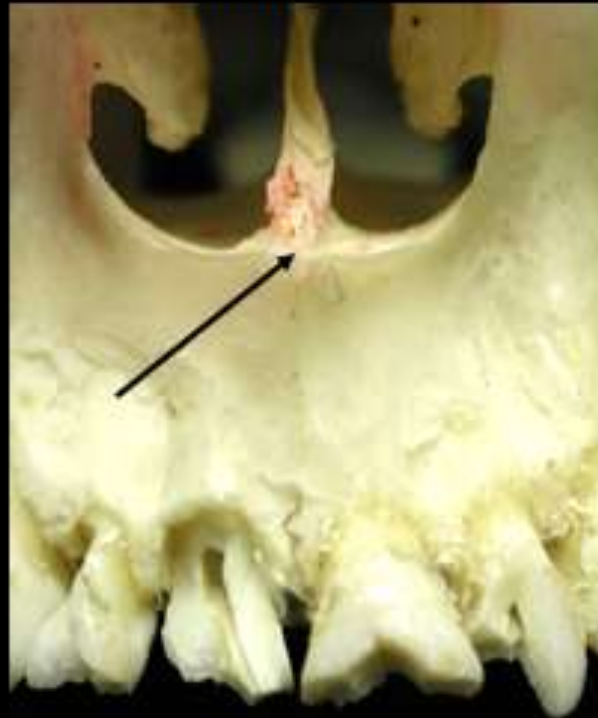






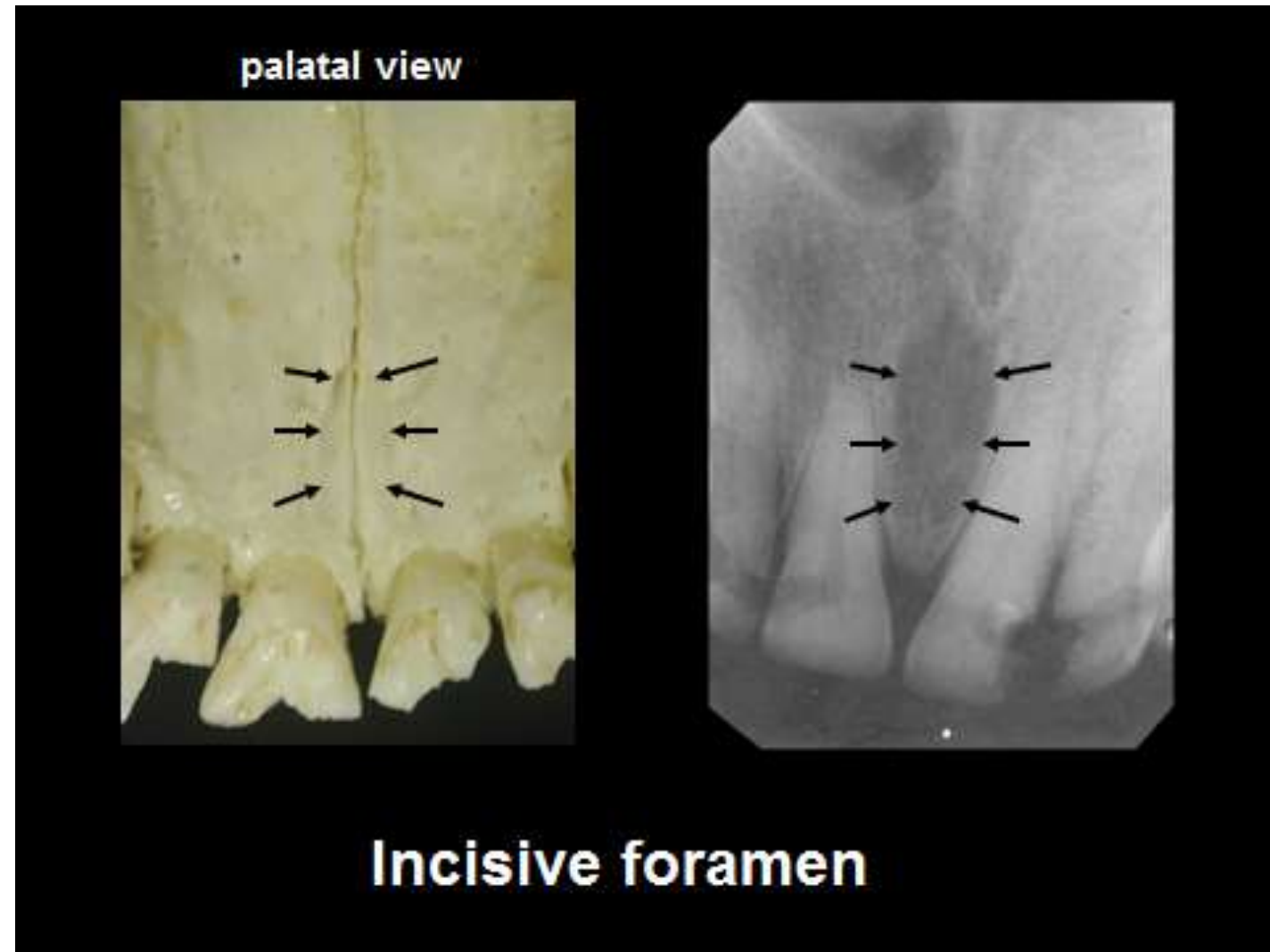


facial view



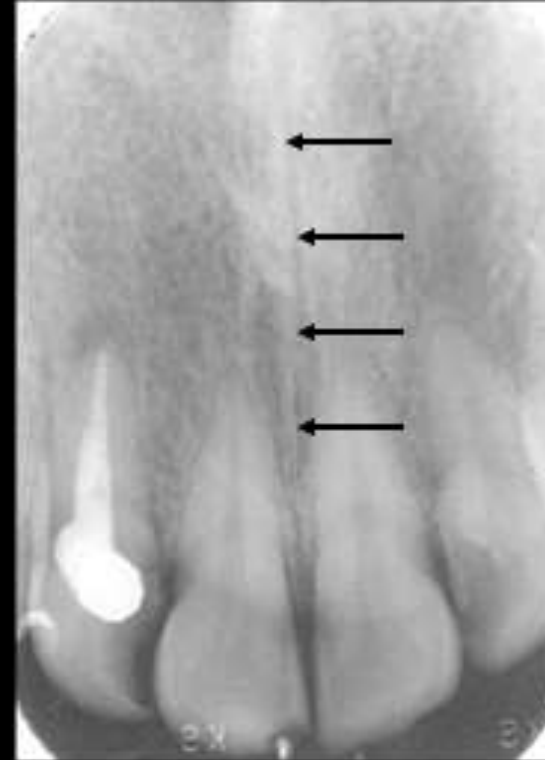
Anterior nasal spine





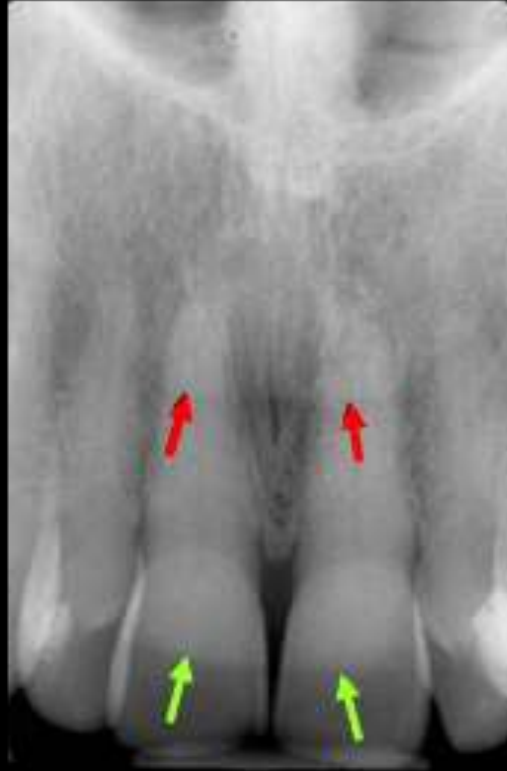


palatal view



Median palatal suture



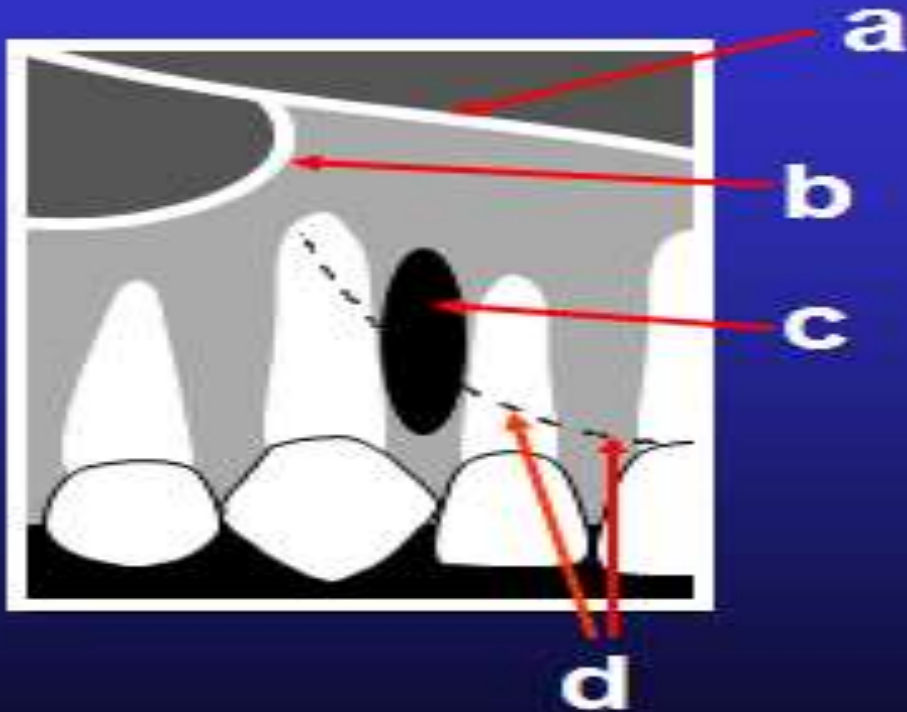


The red arrows point to the soft tissue of the nose.
The green arrows identify the lip line.



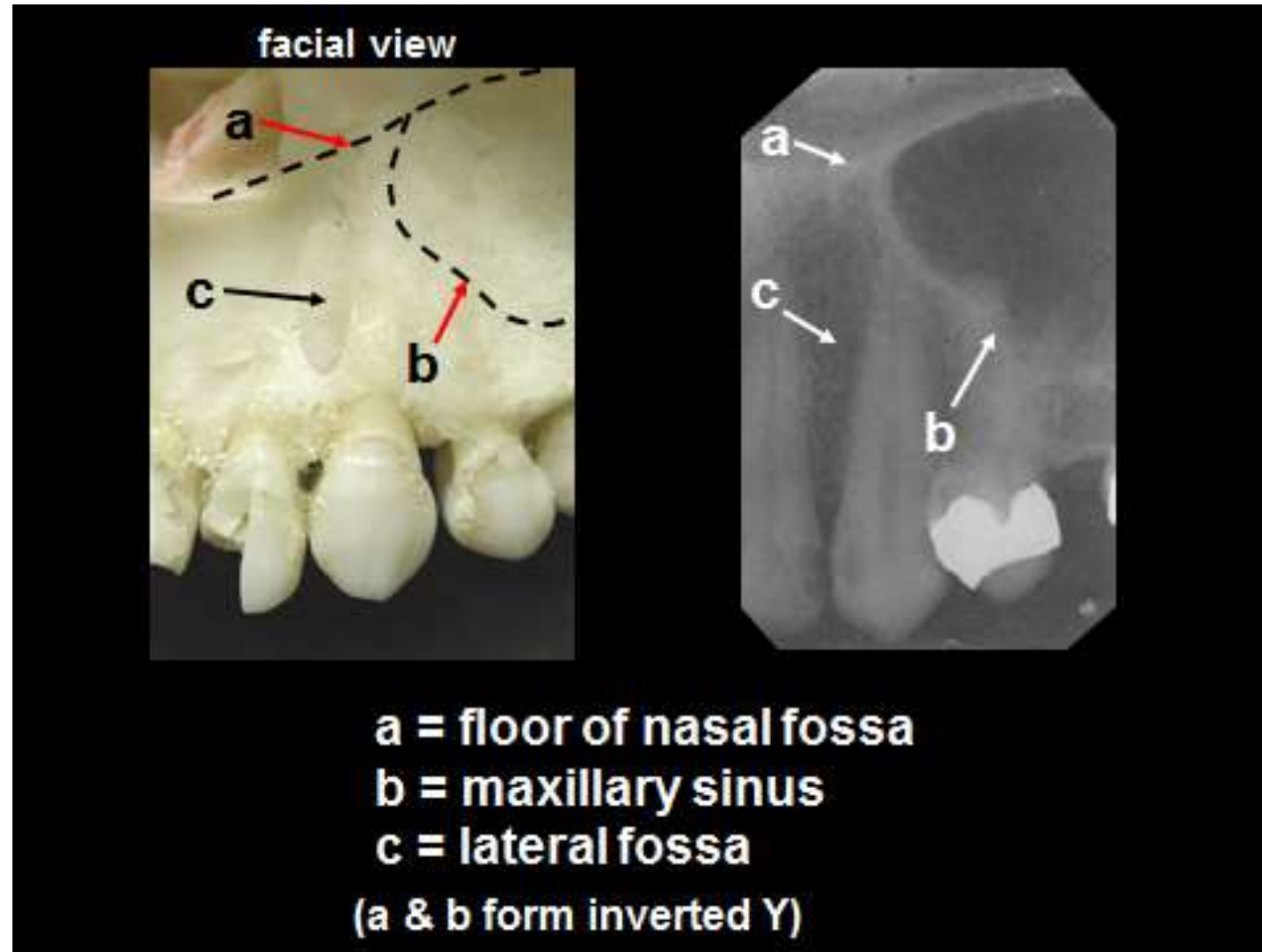


Maxillary Cuspid



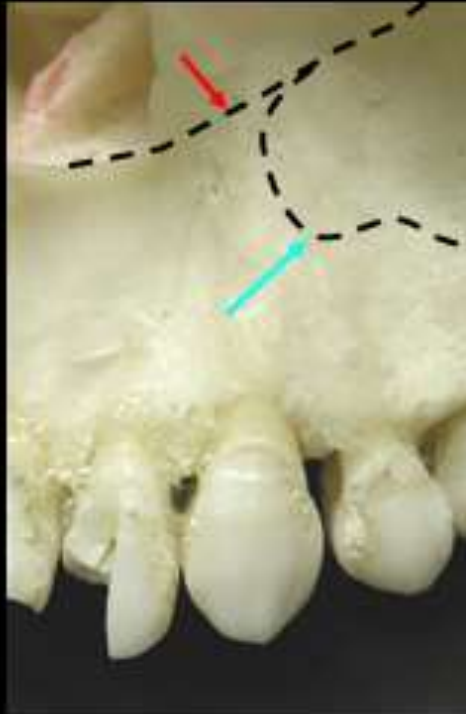
- a = floor of nasal fossa
- b = maxillary sinus
- c = lateral fossa
- d = nose







facial view

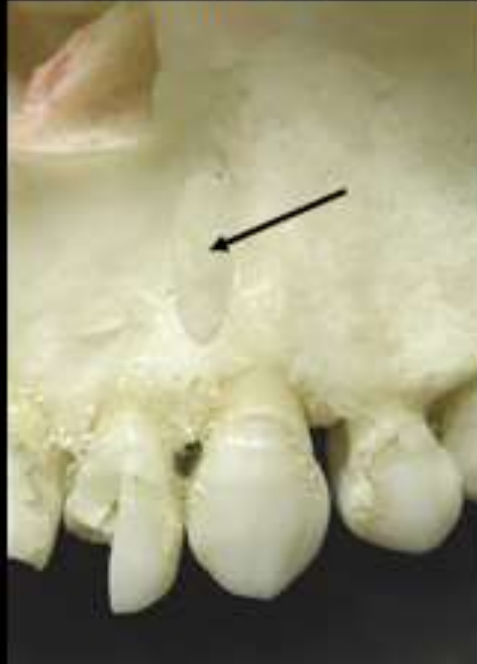


Floor of nasal fossa (red arrows) and anterior border of maxillary sinus (blue arrows), forming the inverted (upside down) Y.





facial view



Lateral fossa. The radiolucency results from a depression above and posterior to the lateral incisor. To help rule out pathology, look for an intact lamina dura surrounding the adjacent teeth.





The maxillary sinus surrounds the root of the canine, which may be misinterpreted as pathology.



The white arrows indicate the floor of the nasal fossa. The maxillary sinus (red arrows) has pneumatized between the 2nd premolar and first molar



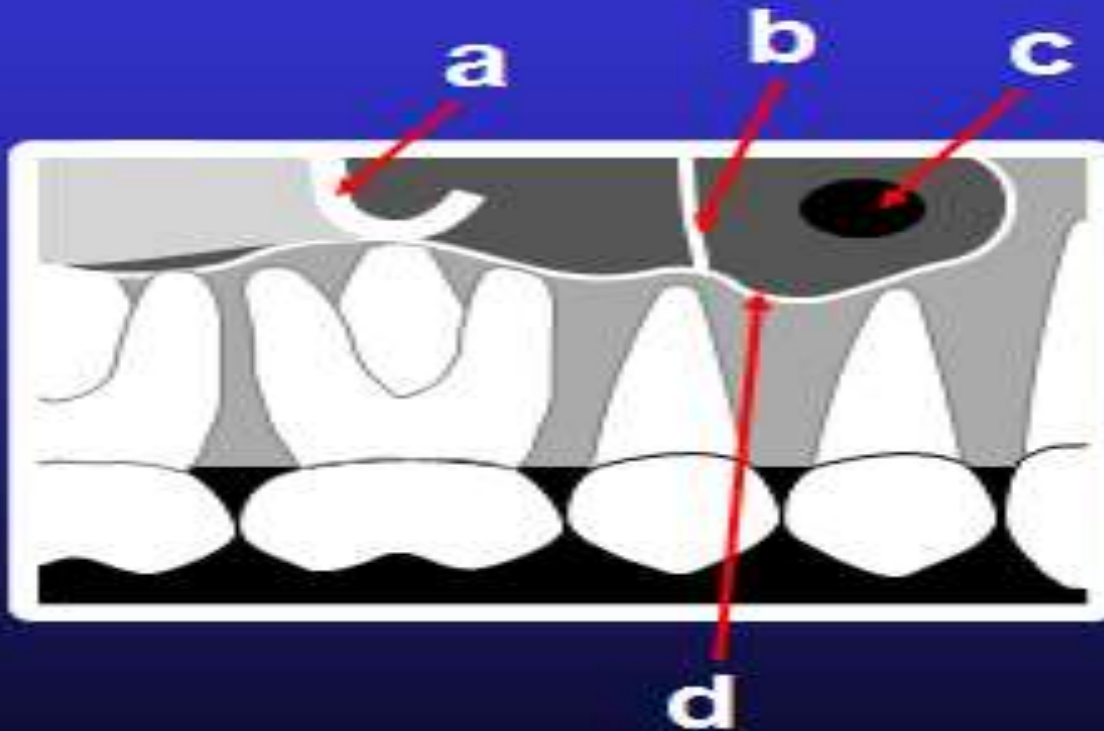


The red arrow identifies the lateral fossa. The pink arrow points to CPP (chronic periapical periodontitis = abscess, granuloma, etc.).





Maxillary Premolar



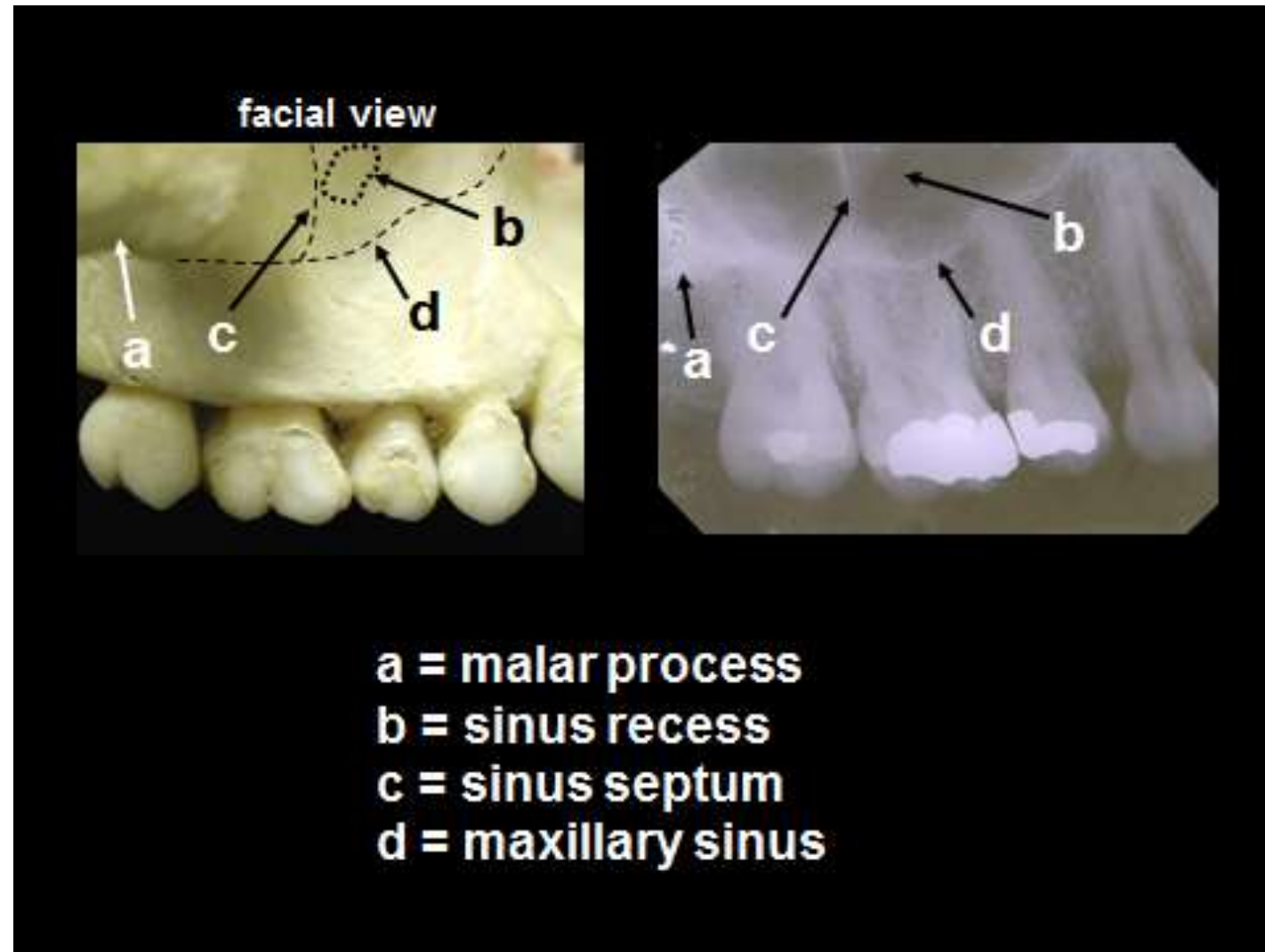
a = malar process

b = sinus septum

c = sinus recess

d = maxillary sinus







facial view

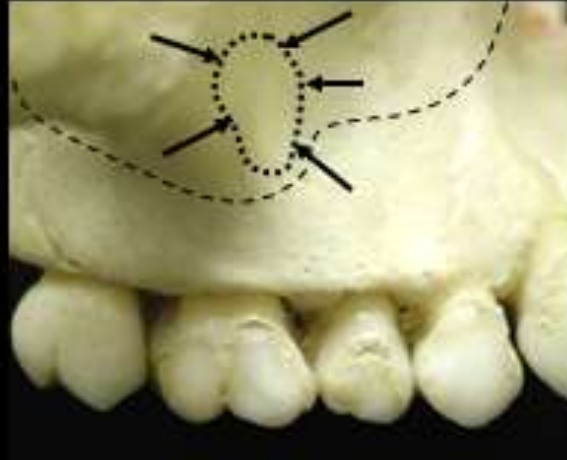


Sinus septum. This septum is composed of folds of cortical bone that arise from the floor and walls of the maxillary sinus, extending several millimeters into the sinus. In rare cases, the septum completely divides the sinus into separate compartments.





facial view



Sinus recess. Increased area of radiolucency caused by outpocketing (localized expansion) of sinus wall. If superimposed over roots, may mimic pathology.





Pneumatization. Expansion of sinus wall into surrounding bone, usually in areas where teeth have been lost prematurely. Increases with age.





Maxillary Molar

a = maxillary tuberosity

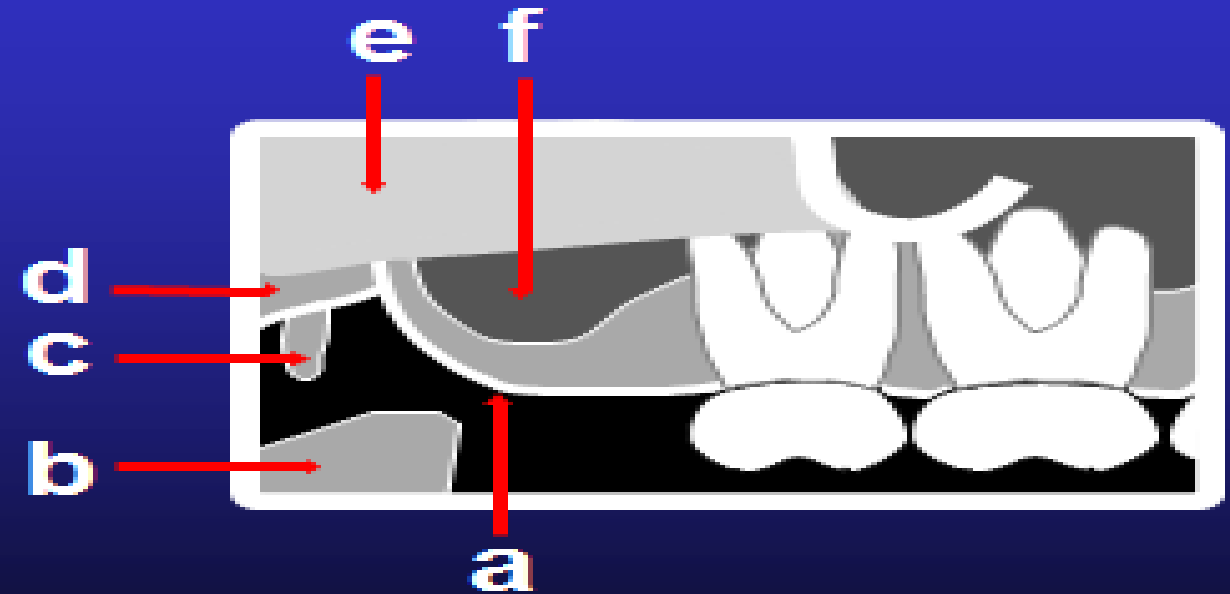
b = coronoid process

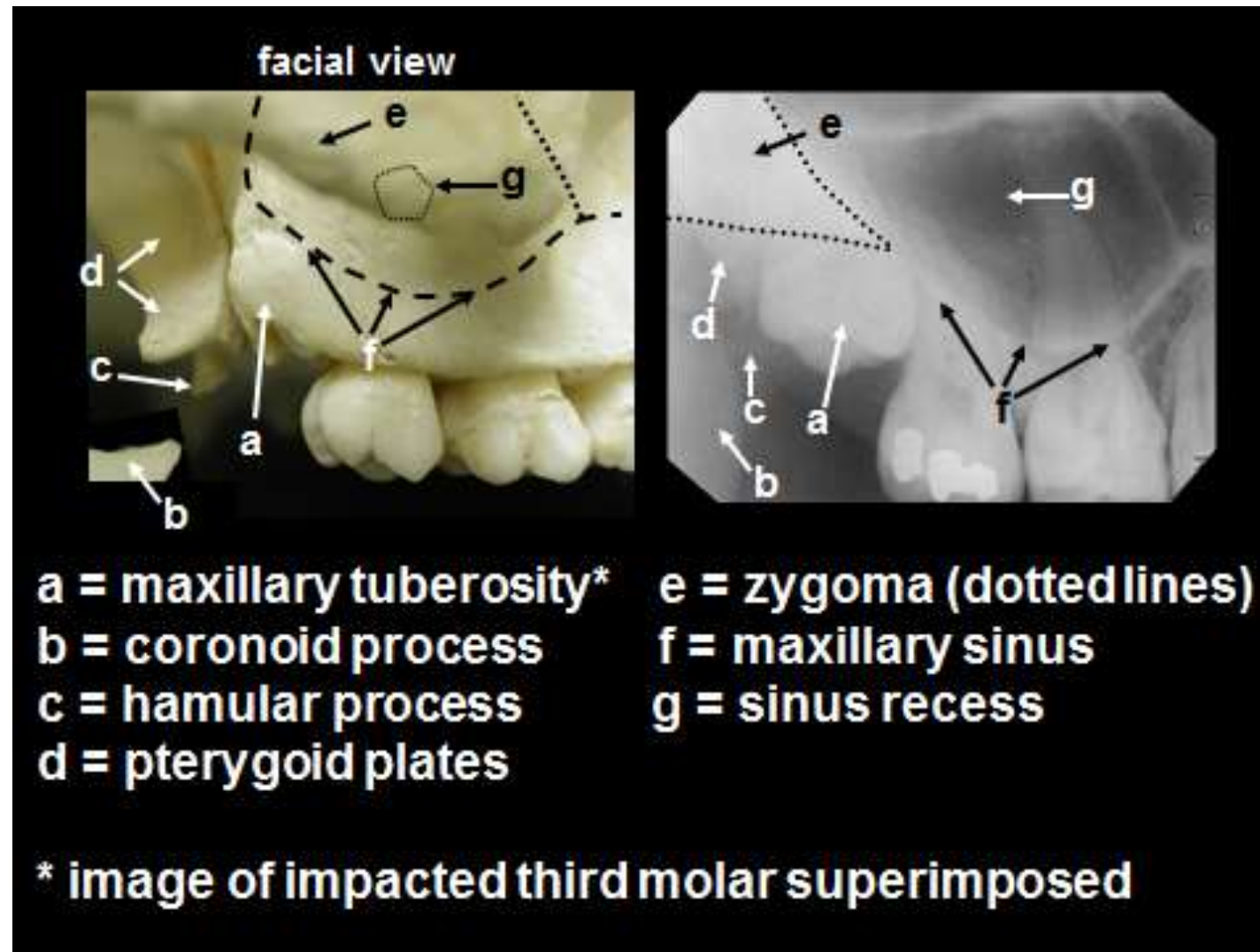
c = hamular process

d = pterygoid plates

e = zygoma

f = maxillary sinus







facial view



Maxillary Tuberosity. The rounded elevation located at the posterior aspect of both sides of the maxilla. Aids in the retention of dentures.





facial view

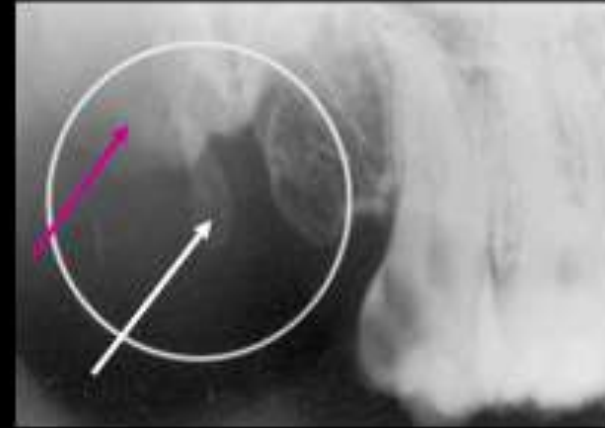


Coronoid process. A mandibular structure sometimes seen on the maxillary molar periapical film when using the bisecting angle technique with finger retention (The mouth is opened wide, moving the coronoid down and forward). Note the supernumerary molar.





facial view

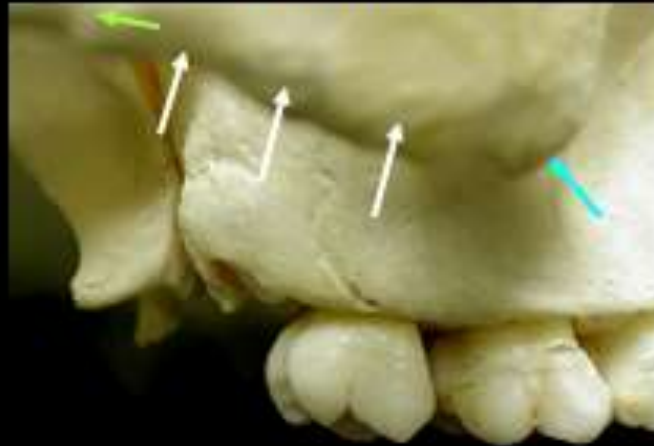


Hamular process (white arrows) and pterygoid plates (purple arrows). The hamular process is an extension of the medial pterygoid plate of the sphenoid bone, positioned just posterior to the maxillary tuberosity.





facial view



Zygomatic (malar) bone/process/arch. The zygomatic bone (white/black arrows) starts in the anterior aspect with the zygomatic process (blue arrow), which has a U-shape. The zygomatic bone extends posteriorly into the zygomatic arch (green arrow).



